



---

MINING DEALER BEST PRACTICE

# Tool for Removing the Teeth Ripper in 16 (G,H,M) & 24 (H,M) Motorgraders

Maintenance and  
Repair

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|     |                                           |   |
|-----|-------------------------------------------|---|
| 1.0 | Introduction.....                         | 1 |
| 2.0 | Best Practice Description.....            | 1 |
| 3.0 | Implementation Steps.....                 | 2 |
| 4.0 | Benefits.....                             | 3 |
| 5.0 | Resources Required .....                  | 3 |
| 6.0 | Supporting Attachments / References ..... | 4 |
| 7.0 | Related Best Practices .....              | 4 |
| 8.0 | Acknowledgements.....                     | 5 |

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices.” Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

September 2014  
1407-2.06-1348

---

## 1.0 Introduction

The ground engaging tool in the Motorgraders must withstand high speeds and high work abrasion on very compacted surfaces. The tips are made of high strength metals and abrasion resistance to provide long service life under tough conditions. However, these undergo wear and should be replaced approximately every 400 to 450 hours. The minimal amount of tip-rippers from motor graders range from 6 to 12 through the 16 to 24 series in all the models, with an approximate weight per tooth up to 10.9 kg. The tips are secured to the shank-ripper through a pin and a retainer; they must fit in well so that the tip does not slip out during operation; These conditions during the disassembly of the tip require a lot of time and also physical effort from the technician by using a pin puller and a hammer to remove them (see Fig. 1). The time length varies depending on the strength of each technician because they need to take rest intervals during the disassembly of each tooth, Moreover, the constant risk due to accidents by blows, bruises, back pain, sore hands, excessive noise, etc. remains because of hammering and poor working postures during the extended working time. (See Figure 1)

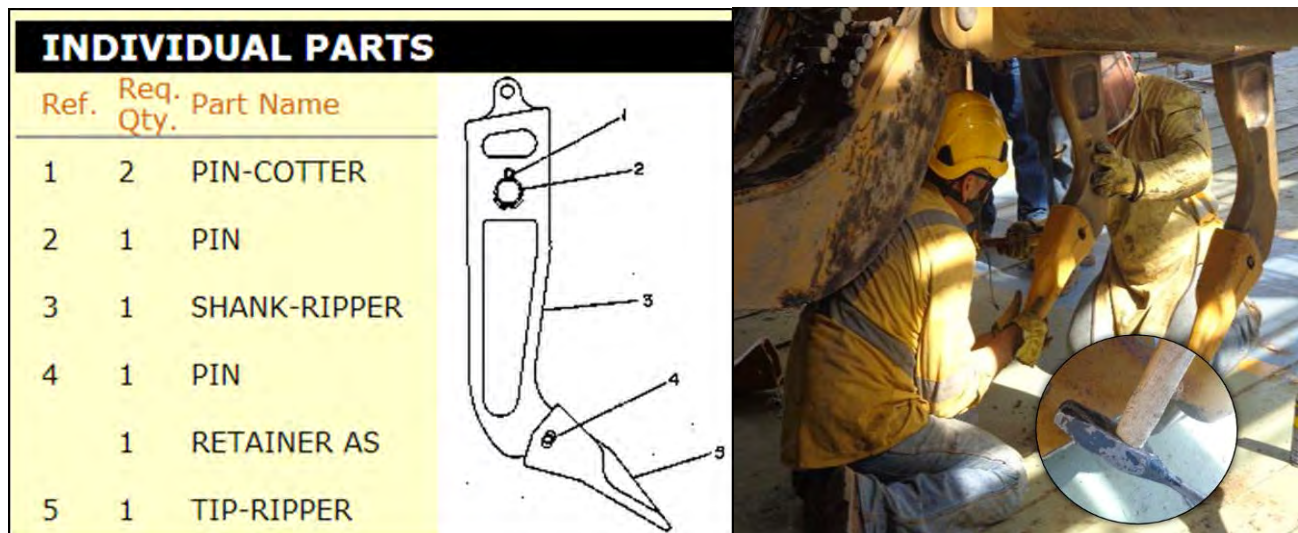


Figure 1. Parts Identification & disassembly process of the tip-rippers

This Best practice proposes using a hydraulic tool that facilitates and streamlines the task of disassembly of the tip ripper regarding the series and aforementioned models, significantly reducing the work time, the physical exhaustion, and accident risk of the personnel.

## 2.0 Best Practice Description

The tool consists of a steel pyramidal liner A572 GR 50 thickness which is inserted into the tip to be disassembled embracing and adjusting it perfectly to its geometry; this is supported through a spring-operated lock that is hooked over the edge of the tip-ripper. It is welded to the liner and it has a ½" thickness reinforced metal armor secured from end to end with 4 bolts (9mm x 140mm) and their nuts and washers. Inside contains a double-acting hydraulic cylinder, 140mm long.

| THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR |                   |                 |                          |
|-----------------------------------------------------------------------------------------------------------------|-------------------|-----------------|--------------------------|
| Tool for Removing the Teeth Ripper in 16 (G,H,M) & 24 (H,M) Motorgraders                                        | DATE<br>2/16/2015 | CHG<br>NO<br>00 | NUMBER<br>1407-2.06-1348 |

The 1" diameter shaft is collinear to the pin axis that secures the tip-ripper being disassembled. The cylinder consists of a quick coupling system for the hose connections to electrohydraulic pump capacity. This creates enough hydraulic pressure to get the rod out of the cylinder and simultaneously displace and remove the pin that secures the tip-ripper to the shank ripper (See Figure 2).

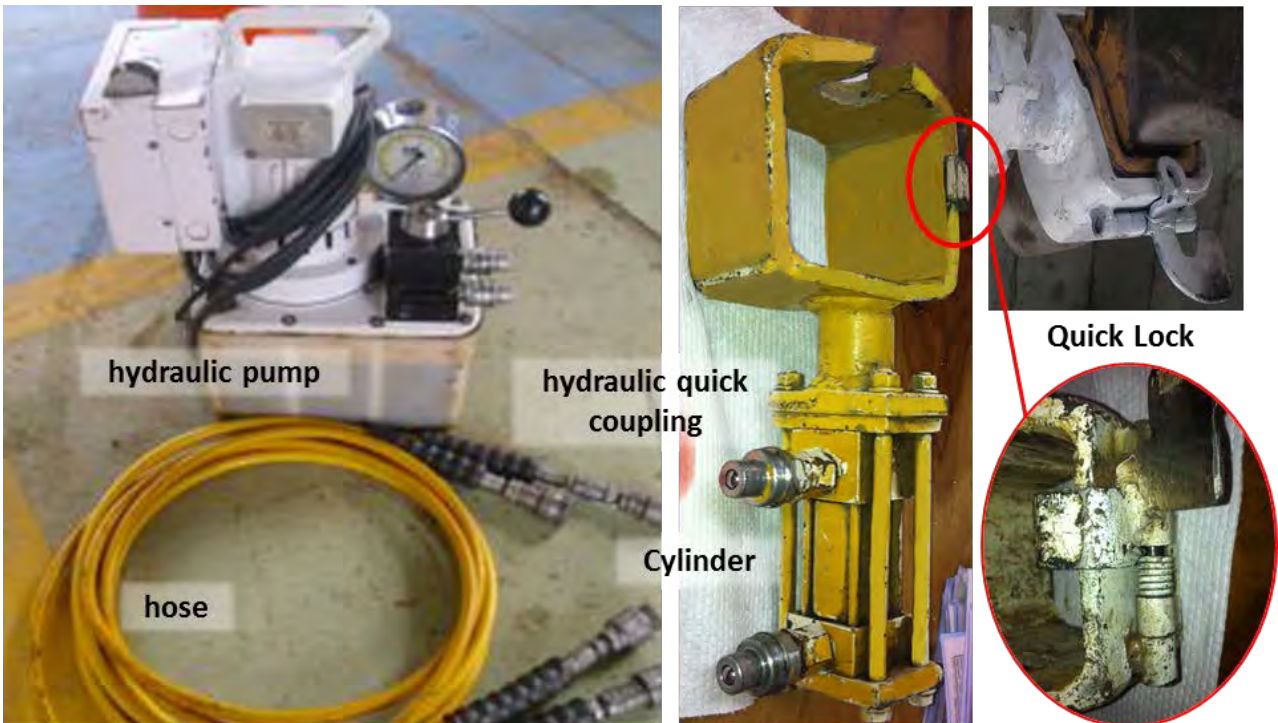


Figure 2.

### 3.0 Implementation Steps

- Make the tool according to the description and attached drawings.
  - Operation (See figure 3)
1. Lock the equipment using the locking tool on the lift cylinder (either of the two).
  2. Check the status of the pump, then proceed to install the electrohydraulic pump and the two hoses (Check the oil level).
  3. Attach the hoses and the pump to the tool, make sure the joints are secured and the fittings are working well.
  4. Install the tool to the tooth. Place the latch to lock the tool to the tooth. Consider the component weight and the body position to work comfortably and safely.
  5. After pulling the pin, hold the tool with both hands to remove it. Remove the tool, clean it and place it in a safe place. CAUTION: If the pin is tight, it can "twang" out of the hole when being pushed out by a hydraulic cylinder. The pin can hit the next shank and bounce back.

| THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR |                   |                 |                          |
|-----------------------------------------------------------------------------------------------------------------|-------------------|-----------------|--------------------------|
| Tool for Removing the Teeth Ripper in 16 (G,H,M) & 24 (H,M) Motorgraders                                        | DATE<br>2/16/2015 | CHG<br>NO<br>00 | NUMBER<br>1407-2.06-1348 |



Figure 3. Top ripper disassembly operation with the new tool.

#### 4.0 Benefits

- Easy manufacturing and operation tool.
- Reduction of accidents due to bruises, back pain, sore hands, excessive noise, etc... due to the elimination of hammering and poor working postures during the extended working time.
- Reduction of the technical personnel for the development of the activity (it is required just one person, previously 2 were required).
- Considerable saving of time in the development of a task (From 4 hours min to 1 hour min approximately) nearly reduce 75%. Thus, it increases the machine availability.

#### 5.0 Resources Required

- Components mentioned in the description:

Hydraulic Pump  
Hydraulic Cylinder (140mm long. The 1" diameter)  
Quick coupling and hoses sets  
Cylinder Plates and Liner for the tip-ripper

- Liner manufacturing drawings and Cylinder plates for the cylinder assembly
- Welder. 8 H/H

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tool for Removing the Teeth Ripper in 16 (G,H,M) & 24  
(H,M) Motorgraders

DATE  
2/16/2015

CHG  
NO  
00

NUMBER  
1407-2.06-1348





## 8.0 Acknowledgements

### **Carlos Mario Herrera Peñaranda**

Project Engineer - CPS & 6 Sigma Black Belt  
GECOLSA - Drummond Ltd Project. - Mina El Descanso  
Email: [carlos\\_herrera@gecolsa.com.co](mailto:carlos_herrera@gecolsa.com.co)

### **Special thanks:**

Jair Barrios (DRUMMOND); Freddy Trespalacios (DRUMMOND); Juan Ávila (DIMANTEC)  
Alcides Rocha (DIMANTEC); Julio Manga (DIMANTEC).

### **Luis Alberto Fernández Ch.**

Development Engineer & BPs Coordinator  
GECOLSA\_CAT Mining Division  
Email: [luisalberto\\_fernandez@gecolsa.com.co](mailto:luisalberto_fernandez@gecolsa.com.co)

Translated by:  
Edgard Romero (GECOLSA, English-Spanish Instructor)

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

|                                                                             |                   |                 |                          |
|-----------------------------------------------------------------------------|-------------------|-----------------|--------------------------|
| Tool for Removing the Teeth Ripper in 16 (G,H,M) & 24<br>(H,M) Motorgraders | DATE<br>2/16/2015 | CHG<br>NO<br>00 | NUMBER<br>1407-2.06-1348 |
|-----------------------------------------------------------------------------|-------------------|-----------------|--------------------------|